

Compendium Addendum Master CLEAN FULL

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CVP Fusion Solutions Compendium - Addendum (Master)

Equation Panel Locked Omega + 4D + EM Integration Expansion Modules P7-P8

Prepared by: Timothy J. Dillon

206 Innovation Inc. Bellevue, Washington

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Canonical Equation Panel (Locked)

Executive Summary

This compendium maps fusion bottlenecks into CVP-native solution modules with concrete algorithms, success metrics, and TEVR proof-bundle artifacts. This revision upgrades the entire set to the locked Dillon Equation notation (c vs c_{eff} ; G_{geom} governance) and adds two long-horizon expansion modules: starlifting and black hole energy harvesting. The core fusion posture remains conservative and engineering-first: stability, recovery, constraints, and replayable evidence.

Problem-by-Problem Solutions (P1-P8)

P1. Real-time stability and disruption avoidance (tearing modes, locked modes, wall contact)

Root cause: Plasma is inherently unstable; many actuators must be coordinated at high rate with sparse/noisy sensors; sim-to-real drift breaks policies.

CVP module: Invariant-Driven Stability State + CVP Safe-Manifold Controller

Solution:

Define invariant state $I(t)$; forecast disruption risk; run constrained MPC/RL hybrid restricted to a safe manifold with recovery gates and hard limits.

Algorithms / implementation:

Predictor: calibrated $p_{\text{hat}}(\text{disruption} | I(t))$ with uncertainty thresholds

Mitigator: constrained MPC or constrained policy optimization with $g(u, I) = \text{True}$

Drift-gate: if recovery tests fail, downgrade to conservative policy

Success metrics:

Avoided disruptions (%), false positives, wall-contact events

Mode amplitude proxy reduction at fixed heating power

Zero constraint violations in claim runs

Recovery tests pass within epsilon

TEVR proof-bundle artifacts:

Frozen setpoints + actuator families; deterministic run-order fingerprints

Replay script raw traces -> metrics -> figures; declared epsilon

Environment/model hashes; calibration chain

P2. Design speed for leak-proof magnetic confinement (stellarator / advanced coil design)

Root cause: Coil optimization is high-dimensional and historically slow; tolerance robustness is often treated late.

CVP module: CVP Design-Invariant Optimizer + Robustness-First Objective

Solution:

Define I_{field} capturing closure/islands/tolerance; optimize $L(I_{\text{field}}) + \lambda R(\text{tolerance})$; TEVR-proof every candidate.

Algorithms / implementation:

Objective: minimize $L(I_{\text{field}}) + \lambda R(\text{tolerance})$

Search: gradient-based where available; otherwise CMA-ES/BayesOpt with constraints

Robustness: Monte Carlo tolerance sweeps as acceptance tests

Success metrics:

Design runtime (speedup), leakage/island constraints satisfied

Tolerance robustness yield under perturbation

Manufacturing feasibility score

TEVR proof-bundle artifacts:

Seed lineage + solver version hashes; geometry files checksummed

Tolerance sweep definitions and thresholds embedded

P3. Edge/divertor heat exhaust and component longevity (SOLPS-ITER class challenges)

Root cause: Heat and particle exhaust concentrate on divertor targets; asymmetries and drifts create strike-point patterns; materials face cycling.

CVP module: CVP Edge/Exhaust Index + Heat-Flux Shaping Controller

Solution:

Define edge invariants I_{edge} and heat-risk $H(t) = \max_x q(x, t)$; train surrogates; control to minimize peak heat while preserving confinement.

Algorithms / implementation:

Surrogate: differentiable $q_{\text{hat}}(x, t | I_{\text{edge}}, u)$ with local recovery constraints

Controller: multi-objective constrained optimization to minimize peak heat flux

Strike-point sweeping schedules that avoid mode excitation

Success metrics:

Peak heat flux reduction (%), distribution uniformity

Time-at-risk above threshold

Confinement proxy maintained (no regression beyond epsilon)

TEVR proof-bundle artifacts:

Model provenance (datasets), training logs, cross-validation

Operational replay of heat-risk decisions vs measured $q(x)$

P4. Alternative confinement (Z-pinch / pulsed pinch systems)

Root cause: Pinch systems face fast instabilities; pulsed timing is unforgiving; diagnostics can be limited.

CVP module: CVP Pulse-Shaping and Instability-Suppression Overlay

Solution:

Treat current drive and timing as waveforms; compute pinch invariants I_{pinch} ; optimize waveform families to avoid instability growth windows.

Algorithms / implementation:

Waveform optimization under hard limits (dI/dt , voltage, arcing, electrode stress)

Band-targeted bounded perturbations to decorrelate unstable modes

Virtual diagnostics to reconstruct missing states with confidence scoring

Success metrics:

Repeatable stable pinch duration; reduced run-to-run variance

Neutron yield per joule (Phase B)

Lower arcing/electrode-damage rate

TEVR proof-bundle artifacts:

Waveform hashes; synchronized timing logs; calibration chain

P5. Virtual diagnostics and sensor scarcity (Diag2Diag-type reconstruction)

Root cause: Commercial fusion cannot afford unlimited sensors; sensors fail/drift; missing channels destabilize control loops.

CVP module: CVP Integrity Plane for Diagnostics (Trust-Weighted Sensing)

Solution:

Fuse physical sensors and virtual diagnostics into trust-weighted measurement; confidence scoring; failover to conservative control when needed.

Algorithms / implementation:

Reconstruction with uncertainty estimates

Invariant consistency checks to flag drift

Failover policy when uncertainty exceeds threshold

Success metrics:

Control stability under sensor dropout; downtime reduction

Reconstruction error vs ground truth; calibration stability

False alarm rate for drift detection

TEVR proof-bundle artifacts:

Dataset partitions + dropout scenarios; uncertainty calibration reports

Replay of reconstructions on fixed test sets

P6. Economic viability: iteration speed, reliability, and cost of operation

Root cause: Shots are expensive; sim/experiment mismatch wastes cycles; partners demand reproducibility; reliability drives OPEX.

CVP module: TEVR Proof Bundles + Micro-Experiment Batching + Robust Optimization

Solution:

Make every claim replayable; increase information per shot with bounded micro-perturbations; optimize for robust performance under uncertainty.

Algorithms / implementation:

TEVR pipeline integrated with data acquisition and analysis

Robust objective: maximize $E[\text{perf}] - \lambda \text{Var}(\text{perf})$ with safety constraints

Shot-time batching to estimate local sensitivities $dP_i/d\theta$ and $dH/d\theta$

Success metrics:

Cost per validated improvement; time-to-decision reduction

Replication success across days/operators

Reduced variance and fewer aborts

TEVR proof-bundle artifacts:

End-to-end provenance: run logs -> raw traces -> analysis -> figures -> signed artifacts

P7. Starlifting: controlled stellar mass extraction (long-horizon infrastructure)

Root cause: Astrophysical-scale extraction must be ultra-low-rate to avoid destabilizing stellar structure; routing and capture require precise field control.

CVP module: CVP Stellar Extraction Overlay (Omega gating + 4D state + EM routing)

Solution:

Model starlifting as controlled thermal outflow plus magnetic routing. Use Omega convergence flags to halt on instability; enforce TEVR proof bundles for energy accounting and extraction stability.

Algorithms / implementation:

Baseline: $v_{\text{esc}} = \sqrt{2GM_{\text{star}}/r_{\text{star}}}$; $E_{\text{spec}} = GM_{\text{star}}/r_{\text{star}}$

Control: localized heating -> plasma fountain; magnetic nozzle routing

Governance: extraction fraction $f \sim \dot{m}/M_{\text{star}}$ bounded (e.g., $\sim 0.001\%/yr$) with recovery gates

Success metrics:

Stable extraction without triggering fragmentation flags

Energy-per-kg accounting consistent with E_{spec} within tolerance

Routing efficiency and containment success

TEVR proof-bundle artifacts:

Frozen setpoints, field maps, extraction logs; deterministic replay; declared epsilon

P8. Black hole energy harvesting: extreme-environment governance (conceptual)

Root cause: Extreme EM/plasma environments demand fail-fast governance and auditable control; near-term practicality varies by mechanism.

CVP module: CVP Extreme-Field Governance Overlay (4D topology + EM control + TEVR)

Solution:

Treat accretion conversion and magnetized spin extraction as controlled routing/topology problems; apply 4D state and Omega stability gating; TEVR governs any operational claim.

Algorithms / implementation:

Baseline: accretion $P \sim \eta \dot{m} c^2$; spin extraction conceptual in magnetized environment

4D state to track topology dynamics and stability margins

Fail-fast: retreat logic on fragmentation flags; strict constraint envelopes

Success metrics:

Stability margin maintenance and safe retreat success

Auditable power/energy accounting

Zero uncontrolled excursions outside constraints

TEVR proof-bundle artifacts:

High consequence proof bundles: full provenance, replay, and acceptance gates

Roadmap and Budget

References (public sources)

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